

DAVE stands for 'Data Analysis and Verification/Visualization/Validation Environment' and its source code is publicly available on GitLab.

The free software enables very large volumes of data, such as those produced by modern sensor systems, to be processed, analysed and visually displayed very rapidly, so that researchers can optimise their measurement systems interactively.

A short introduction given in the readme of the repository, says, "DAVE is a MATLAB-based toolbox for the evaluation of, mainly, cyclic sensor signals. It



focuses on cycle-based raw data preprocessing, graphical feature extraction, and data annotation, but also provides commonly used machine learning methods (with cycle-specific extensions) to develop data-driven models. This allows for a sleek workflow from

start to finish without having to change to third-party tools."

Instead of relying on a conventional and time-consuming trial and error approach, the new software effectively asks the question 'What happens when...'.

"Whenever we use our gas sensors to measure air pollutants, we are faced with the same old problem of analysing vast volumes of data and of recognising signal patterns. If we want to continue to make our sensors more sensitive and more selective, we need to know whether very fine modifications to the sensors themselves and to the analysis actually bring about the desired improvements in sensitivity and selectivity. But there are countless ways in which sensors can be modified. We want to be able to identify the best paths as rapidly as possible, or, equally, to quickly detect and reject the unproductive paths," explained Professor Schütze.

"The software makes use of machine learning methodologies and enables us to identify patterns rapidly, to evaluate data cleanly and to visualise our results," Schütze said.

The researchers are now publishing their software tool under a copyleft licence, under which, any adaptations of the original work, such as changes or enhancements are also bound by the same licence that covers the original work.

"Anyone may use the open source software, provided that when results are published, the authors make reference to DAVE," asserted Schütze.

The researchers claim that any amount of sensor data can be processed with this software tool.

"DAVE is the opposite of a black box. The software makes the calculations completely transparent. It shows the user that when they alter a particular parameter, it has a specific identifiable consequence. The visualisation modules in DAVE also make it easier to optimise a measurement system. The user can run through, test out and visualise different variants, and that helps the user find the most promising variants quickly and efficiently," explained Manuel Bastuck, a research assistant in the Measurement Technology Lab and the developer of the software.

Prof. Schütze added, "Using DAVE as a tool, we were able to rapidly achieve some widely acclaimed results in the field of condition monitoring in 'Industry 4.0' applications. The results not only helped to solve the measurement problem itself, but also to configure the measuring system more simply and more cost-effectively."

OpenSDS builds open autonomous data platform to address data management challenges

All technology trends are data-centric and data intrinsic today. As a result, organisations are facing challenges in managing their data across heterogeneous storage vendors, different platforms and across multiple clouds.

OpenSDS, a Linux Foundation project, is working to build an open data autonomous platform to address these challenges. For this, it is integrating all disparate data services, such as data migration, replication, life cycle, protection, etc, with artificial intelligence and machine learning. The platform is self-governed and intelligent.

OpenSDS is an ecosystem consisting of vendors, end users, solutions providers, technologists and developers from across the globe.

It has members from IBM, Fujitsu, Huawei, Intel, Western Digital, Dell-EMC, Hitachi, LINBIT, Vodafone, KPN, NTT Communications, China Unicom, Yahoo! Japan, Toyota Infotech, GMO, IJJ, Click2Cloud, Oregon State University, and many others.

The OpenSDS community is also sponsoring and conducting an event called 'The Open Data Autonomy Mini Summit' at KubeCon Europe 2019 in Barcelona, Spain, in May this year. This all-day event will focus on data and storage challenges and solutions in the cloud native era. The mini summit will have a mix of technical sessions, end user case studies, interactive workshops, demos and discussions on the roadmap going forward. A panel of experts will speak on data management challenges and how to collaborate to build solutions.

KubeCon, the Cloud Native Computing Foundation's flagship conference, is one of the biggest of its kind in the world. The event, to be held from May 20-23, will bring together adopters and technologists from leading open source and cloud native communities in Barcelona.